

PHY 408 Final Project - Time-Series Analysis of Confirmed Exoplanet Light Curves

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1 Introduction

The search for exoplanets has become increasingly relied on the analysis of time series, by examining stellar light curves, we are able to identify periodic brightness dips associated with planetary transits. Transits occur when a planet passes in front of its host star, blocking a portion of the starlight, thus producing repeating characteristic patterns in the observed flux data. These periodic patterns can be subtle and filled with noise, due to the effects of stellar variability, thus require careful signal processing to detect. The aim of this project is to attempt estimating orbital periods of selected exoplanets using time-series techniques, and evaluating the effectiveness of various filtering, frequency-domain methods, such as seasonal decomposition, Hann and Savitzky-Golay filtering, Fast Fourier Transform (FFT), auto-correlation, and phase folding, in recovering transit signals under varying noise conditions, using the light curve data^[1] (normalized stellar flux measurements) from NASA's Kepler space telescope.

2 Methods

Initiating our analysis, the ExoTrain data was loaded in to Jupyterhub, then sorted and assigned by types, finding a total of 37 exoplanets by their labels through reading over the indexes in the data dictionary. Groups of these exoplanets' raw flux data were then plotted against time steps (Figure.1) and examined for each of their unique features. A NaN check was also performed to ensure that there are no missing values through the 'np.isnan' built-in function.

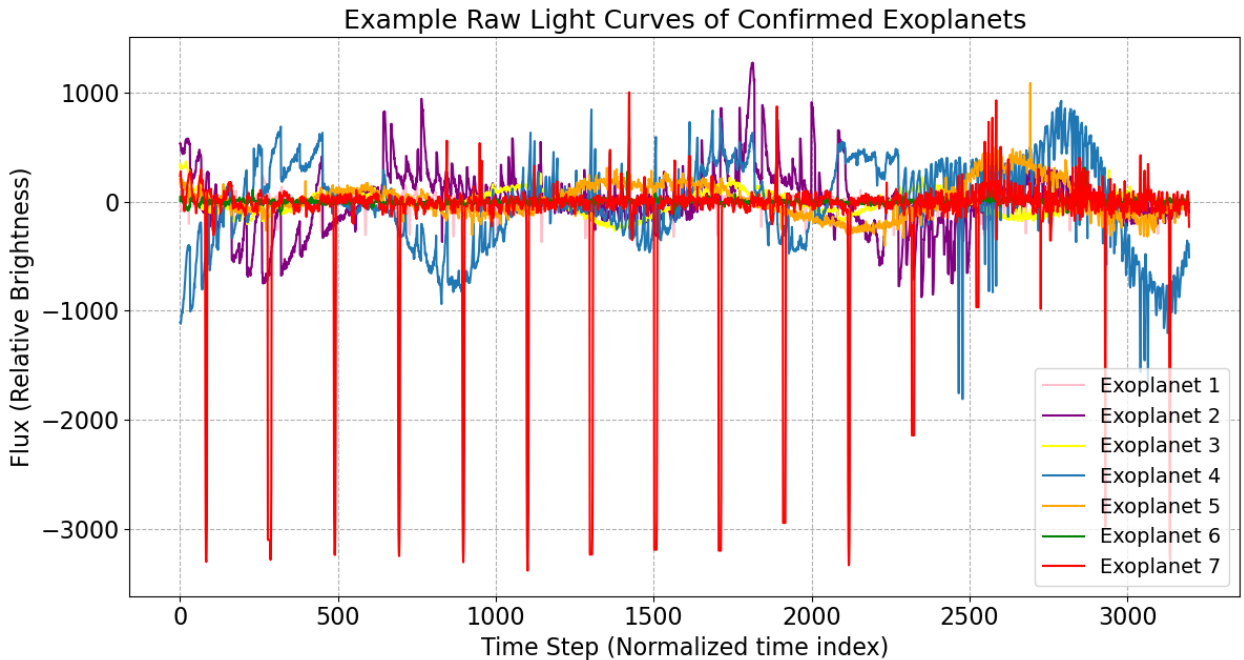


Figure 1: The example raw light curves, plotted with Flux against Time Step, where each time step represents 30 minutes (0.5 hours) color coded on the same figure for examination. Exoplanets 4,5,6,7 were picked for analysis to diversify applications of time-series tools.

Based on these features, four specific exoplanets were selected to increase the variety of

time-series analysis applications. These four relative brightness plots were then normalized by subtracting the data with mean divided by its standard deviation, and plotted again to enhance visualization for trends and seasonality. Further more, we continued to prepare for the analysis by applying FFT to the raw light curves, plotting out four of their according figures (Figure.2) as an initial diagnostic to assess the presence and clarity of signals in each dataset, using the FFT built-ins.

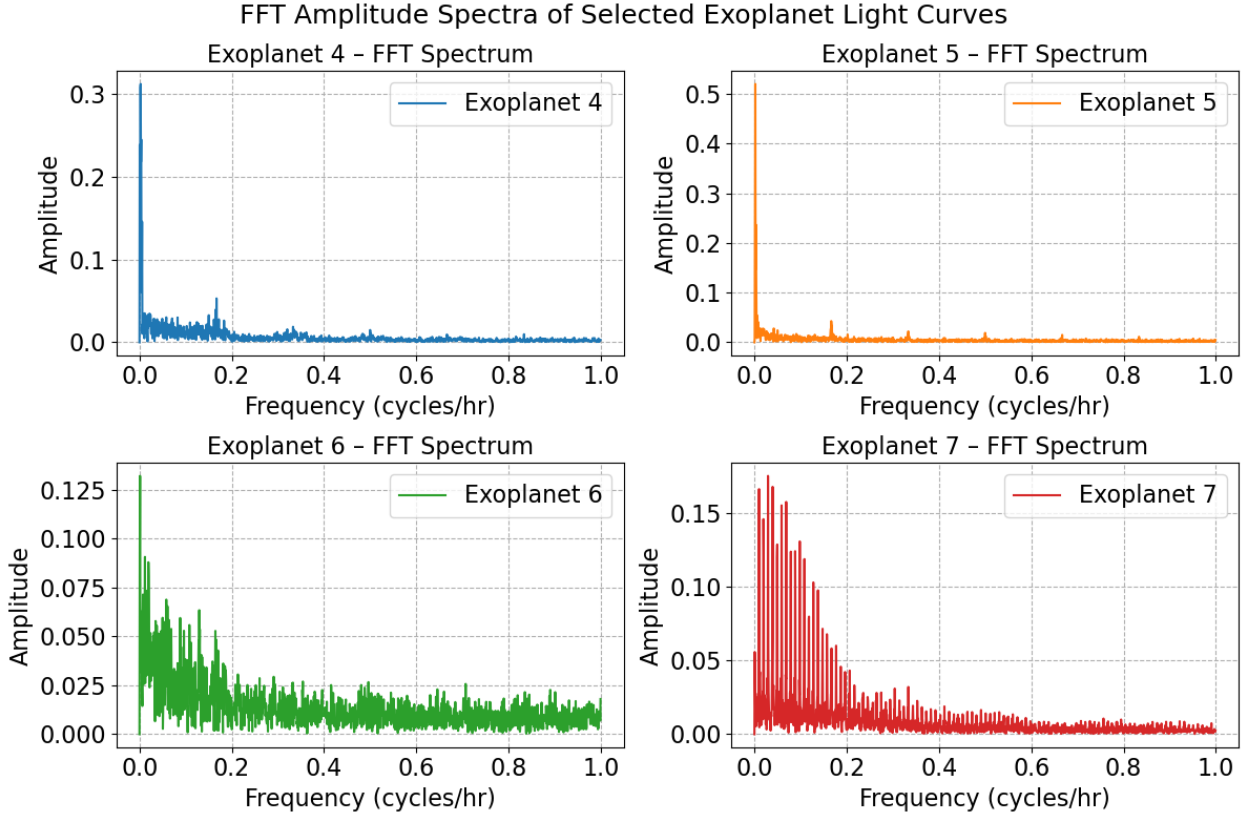


Figure 2: The FFT spectrums of four selected exoplanets, with their amplitude plotted against frequency in [cycles/hr], where exos 4 and 5 presents the same strong zero frequency peak, while exo 6 shows no distinct peaks expect the zero peak. Exo 7 in red shows a harmonic-dominated spectrum, with multiple sharp, evenly spaced peaks.

Examining Figure.2 above, exoplanets 4 and 5 show distinct peaks near low frequencies, leading to a strong periodicity, while exoplanet 7 showed a dense harmonic structure, suggesting the presence of sharp narrow features that could lead to misinterpretation. On the other hand, exoplanet 6 exhibited no dominate peaks and a broad spectrum in the frequency domain, prompting us to consider more adaptive, none-stationary methods.

As these initial spectra helped guide the choice of pre-processing, it was decided that seasonal decomposition could be an applicable start for exos 4 and 5, separating the trend and seasonal component in attempt to de-trend the data, other de-trending attempts were also applied, such as the hann window and the low pass filter, followed by performing a FFT in extracting the dominant frequencies, thus obtaining a period estimate. Exoplanet 6 on the other hand, has no clear periodicity while containing multiple noise bursts, which suggested that we start with a Savitzky-Golay filter, smoothing the data without erasing spikes. Such a

filter performs just like the hann window or the box car filter, fitting a low-degree polynomial to a sliding window, but preserving more local structures. As the above being done, a STFT spectrogram was then applied, in attempt to spot a potential trend or transit pattern to help us estimate the period. In terms of exoplanet 7, we applied a hann window de-trending, followed with a FFT estimation of the period just like with exos 4 and 5. However, the reliability of this method was further examined, since a misalignment in time-domain period was noticed, and that the light curve had such stable and constant dips. By applying phase folding, peak detection, range controlled FFT, we were able to uncover the true spacing for estimating orbital periods, and adapting our methods for a more accurate result.

3 Results and Analysis

3.1 Exoplanets 4 & 5

Analyzing exoplanets 4 and 5 together, as they both exhibited visibly repeating dips, gaining amplitudes in flux as time step progresses. Examining their normalized light curves and FFT spectrums (Figures 1 & 2), it was observed that exo 5 has a clearer, more stable transit pattern compared to exo 4. Thus, while applying seasonal decomposition to both of the exoplanet light curves (Exo 5: Figure.3; Exo 4: App.Figure.7), exo 4 was further analyzed by applying the hann window, not for truncating the data, but smoothing the signal by convolution, acting as a weighted moving average filter.

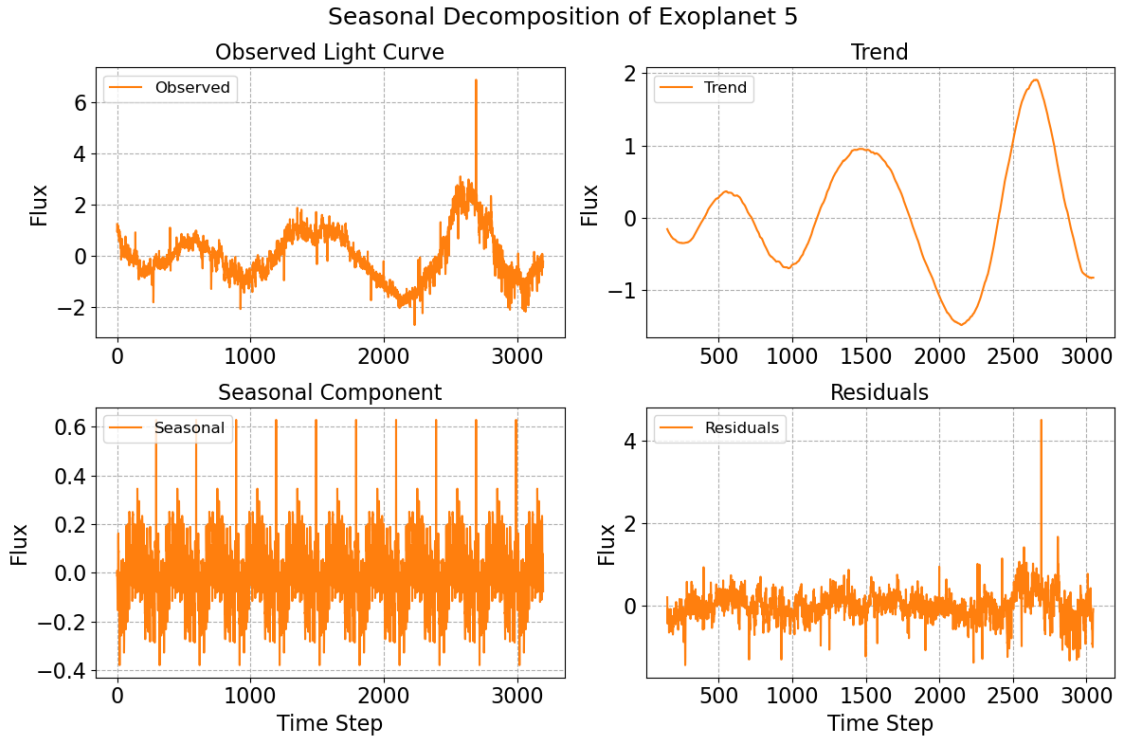


Figure 3: The seasonal decomposition plots of exoplanet 5, containing subplots of the original light curve(upper-left), extracted trend(upper-right), the extracted seasonal component(lower-left), and the residuals(lower-right; all plotted with flux against time step.)

Performing the seasonal decomposition as in Figure.3, one may assume that the trend does not in fact lead to the transits, a more probable assumption is that the stable seasonal

components extracted serves as a better representation of the orbital motions. Thus, an estimation of the transit period for exoplanet 5 was made from taking the FFT of the seasonal component (Figure.4.bottom), by taking the inverses of dominate frequencies, yielding a result of 6.01[hours]. This result was then further confirmed with another estimate from the FFT of its Hann window smoothed light curve, and a reconstructed smoothed flux plot in the time domain (App.Figure.8&9).

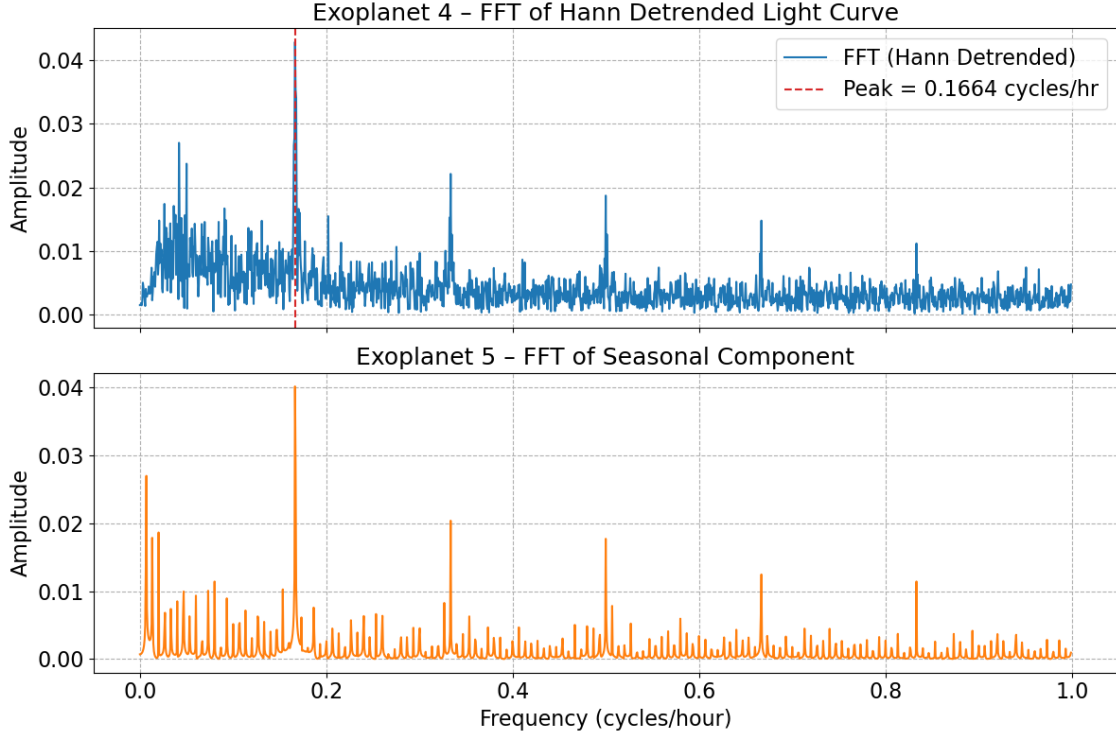


Figure 4: **Top:** the fast Fourier transform of Hann window de-trended light curve of exoplanet 4; **Bottom:** fast Fourier transform of the seasonal component light curve for exoplanet 5. Both plots exhibit apparent dominant peaks in the frequency domain.

On the other hand, exoplanet 4 requires more careful handling as the level of noise has significantly increased. Attempting a similar seasonal decomposition, its seasonality along with its FFT plot provides us with inefficient details to estimate the period from dominant frequency peaks. Therefore, a Hann window was applied to smooth the data, outputting a more eligible FFT plot (Figure.4.top). The process then repeats steps for exo 5, where inverses of dominant frequencies were used to estimate an orbiting period of 5.99[hours]. Notably, when applied with a low pass filter (Butterworth), the de-trended FFT provides similar insights, but with slightly deviated initial noise (App.Figure.10).

3.2 Exoplanet 6

Having the lowest relative brightness in our dataset, exoplanet 6 presented a unique challenge in our analysis, it was only visible after normalizing as in Figure.2, displaying irregular variability and noisy behavior. Furthermore, it is also the planet with the most relative noise, sudden fluctuations, and least visible trend. Initializing, de-trending techniques were applied to isolate potential periodic structure, a Savitzky-Golay filter was applied in hope to

extract smooth trends while preserving the sharper peaks. However, its FFT was broad and lacked dominate frequency peaks (App.Figure.11), suggesting the absence of a stable orbital signature. Although possible minor periodic may exist, the extremely small flux provides little to no help in such relatively large noise, thus, it was decided that we assume exoplanet 6 does not yield interpretable periodicity for our analysis.

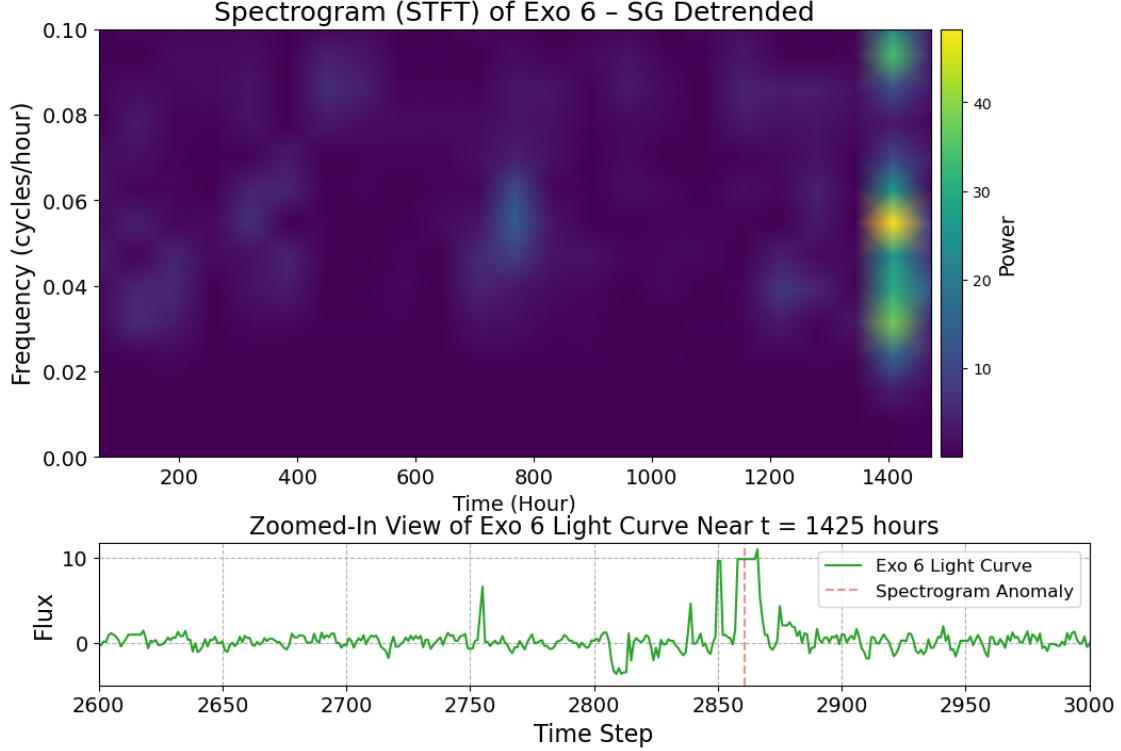


Figure 5: **Top:** The spectrogram of Exoplanet 6, plotted with frequency in cycles/hour against Time in hours, with powers in each location label by a color bar on the right; **Bottom:** The zoomed in view that checks for the peak in light curve, with the same flux vs time step plot, the approximate peak area is marked with a vertical red dash line.

Never the less, we are still able to extract useful information from other aspects of the detrended data, gaining insights to this exoplanet. Investigating the only peak left within the Savitzky-Golay trend, one may suspect that an unusual event taking place, although other deviations such as peaks and dips in the time domain are present, such a broader peak among the rest is still worth investigating. Attempting a time-frequency analysis with the Short-Time-Fourier-Transform, in finding transient features that might not appear in the global FFT, a spectrogram was plotted (Figure.5.Top), presenting a broadband power spike near time step 2850, which matches our peak location. To correlate this spectral anomaly, we zoomed in on the time domain near $t=1425$ [hours](≈ 2850 time steps), and revealed a isolated flux deviation, thus concluded the possibility of a high-energy anomaly, such as stellar flare, or noise artifact rather than a planetary transit.

3.3 Exoplanet 7

Examine the light curve of exoplanet 7 (Figure.1), one can easily see deep, sharp dips recurring at periodically intervals, with rest of the data being relatively flat. This strongly

suggests the presence of a periodic transit signal, where unlike the pervious exoplanets, this one we can visually identify without further cleaning of data. Regardless, a Hann window filter was still applied to the raw light curve to smooth the data; we see that in this case, filtering does not aid much in refining our analysis, since little to none observed difference between the raw (Figure.2.lower-right) and post filtered (Figure.6.top) FFT. The first thought at this point was then, once again, estimating the period with dominant frequency from the FFT, which was printed as 25.37[hours].

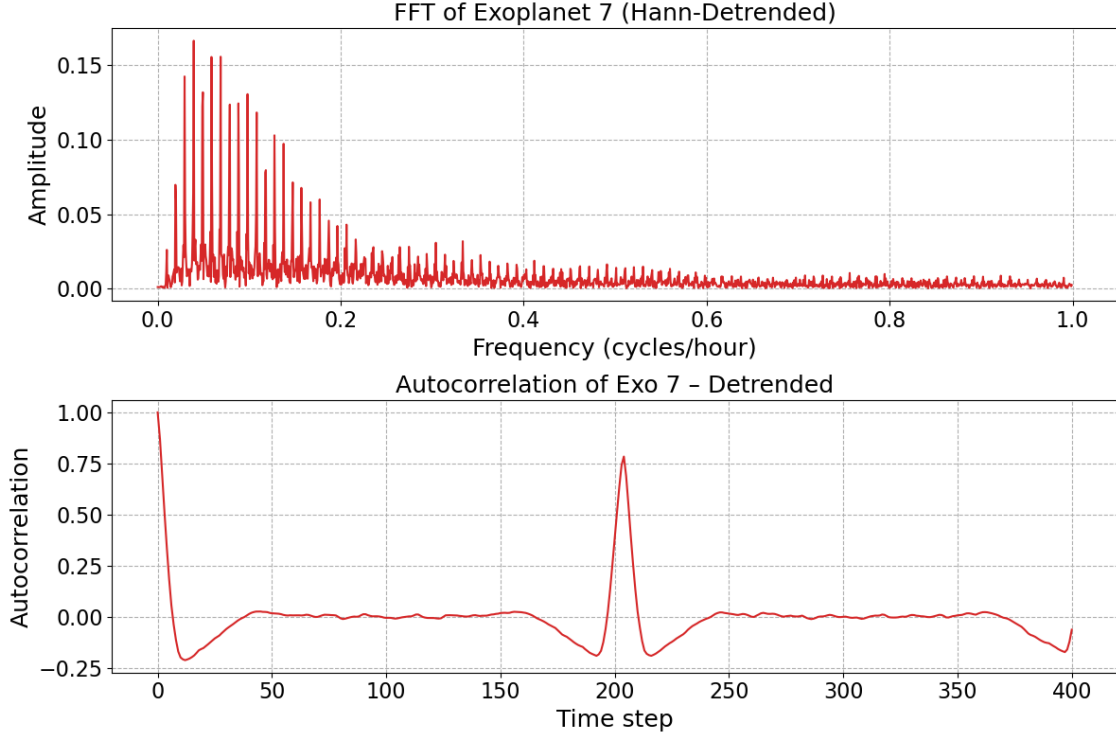


Figure 6: **Top:** The FFT plot after the Hann window filter has been applied, with strong initial amplitudes; **Bottom:** The Auto-correlation of exoplanet 7, suggesting a peak around time step of 200 (≈ 100), aligning with the raw flux plot.

One may immediately realize this serves as a wrong estimate, simply from examining the gaps between our assumed period peaks. An attempt in solving this issue is then, to apply auto-correction, which provides a complementary time-domain assessment of periodicity, measuring similarities between signals and other lagged versions. The autocorrelation plot (Figure.6.bottom) revealed a peak around a lag of 200 time steps(100 hours), which is in close match with the visuals of raw flux plot in Figure.1. To back up such finding, a simple peak to peak measuring from peak detection was applied to confirm this period(App.Figure.12), followed by a FFT estimation method limited to low ranges in the frequency domain,(App.Figure.13), suggesting a transit period of 101.77 and 106.57 hours accordingly. To even further confirm this visually, a phase-folded light curve plot was created, with promising results(App.Figure.14).

4 Data

As mentioned above, the analysis in this report is based on the ExoTrain dataset^[1], which contains stellar light curve data from the NASA Kepler Space Telescope. The data is organized as a time series of normalized flux values recorded at 30-minute intervals (0.5 hours per time step), with each star represented by a 1D array of relative brightness measurements a little over 3000 time steps. As a part of the pre-processing, 37 entries were labeled as exoplanets, which were identified by reading through the data dictionary. Their corresponding light curves were then extracted using these dictionary labels, with each checked for missing values using `'np.isnan'`, making sure none missing values were existing. The dataset was retrieved from Kaggle (See reference) as a CSV file, then loaded into a JupyterHub environment for processing.

5 Discussion and Conclusions

One of the critical observations in analyzing exoplanets 4 and 5 comes down to classifying useful data for estimating the transit periods. Revisiting the seasonal decomposition plot for exo 5 as in figure.3, the extracted trend could be easily mistaken as transit patterns, one may argue that the different locations and amplitudes of flux are merely due to the orbiting angles changing. However, in an astronomical perspective, for such an object orbiting at the according time scale, it is very unlikely for the flux to have such drastic changes, tripling the amplitudes in flux within 3 rotations, therefore, we conclude that a period analyzed from the seasonal component would be more fitful. In contrast exoplanet 4 was considered to have similar characteristics, but holding more noise in its data, a Hann window was used to smooth the data, as a weighted moving average filter. Although we were able to identify the dominant peaks, noise is still present in the frequency domain, suggesting more potential methods to enhance this estimate, such as applying more filters or applying the autocorrelation in the time domain. Validating the results by reconstructing the signal in time domain, we were able to conclude that for exoplanets such as 4 and 5, the underlying periods need to be computed from the smaller apparent trends (larger peaks in frequency domain). Furthermore, in examining the residuals from both of the seasonal decomposition plots (Figure.3 & App.Figure.7), we are able to assume a potential high energy event around 2750 time steps for exo 5, and a blockage in flux around 2500 time step for exo 4.

In terms of Exoplanet 6, a Savitzky-Golay filter was applied to its extremely noisy light curve with no obvious periodicity, such a filter is more suitable because a transient event was suspected in the raw data, while traditional smoothing methods tend to flatten sharp features, the Savitzky-Golay provides a better attempt in smoothing data without overlooking the large peak. De-trending and plotting the signals in frequency domain as in App.Figure.11, no obvious dominant frequencies were found, combining with the visual examinations from the normalized light curve, it was decided that the data set either had no periodic motion that is useful for analysis, or that the actual changes in flux were too small to analyze compared to the existing noise. From here, it then makes sense to examine the suspected transient event instead, as above results introduced with Figure.5. Although we were not able to estimate a reasonable period, the analysis of this exoplanet still proves useful in understanding the utilization of time series tools in astronomical contexts.

In contrast to exoplanets 4 and 5, exoplanet 7 exhibited much clearer flux patterns, to the point where we are able to take them directly for evaluation of orbiting period. Notably, The trend presented by exoplanet 7 has no changes in amplitude nor in periodic lengths, the dips in flux also acts as a very dominant element compared to other nearly flat data in the light curve. With that being said, we still applied the same frequency domain estimation to its data, this time, finding a misalignment with our assumption, this was due to the FFT estimation picking up the lower flux amplitude signals. However, this result does not reject our method, Although other methods such as autocorrelation or peak detection could also be applied in search of the period, we can still estimate period from the frequency domain, by adding a limit to ignore the higher frequency data (App.Figure.13). As a result, all methods out put an estimation that aligns with our visual assumption, with a orbital period of around 200 time steps, 100 hours. To further confirm this result, a phase-folded light curve was plotted with period ≈ 101.77 hours, returning nearly identical curves folded on to each other. With it, we have exemplified with exoplanet 7, that deeper reasoning are required when analyzing real-world astronomical data with time series techniques. Examining the data in multiple domains, integrating visual, spectral, and statistical evidence, lead us in arriving at a confident and physically reasonable estimate.

As a summery, The project presented several challenges, such as the difficulty of extracting reliable periodic signals from noisy light curves, and deciding the useful components of the signal. Other approaches that can potentially make these challenges easier, are the Gaussian process regression, or empirical mode decomposition, however, more time would be needed to test and compare. Another implication that can significantly improve our methods would be to apply machine learning, as ML models can automatically learn to recognize patterns in large datasets. The Exoplanet Detection by Senad Palislamovic open source project^[2] demonstrated a supervised learning approach using deep neural networks trained on the same Kepler light curve dataset. On the other hand, insights from this analysis are shown to be applicable to exoplanet analysis in astronomy fields, specifically in identifying physically meaningful and reasonable data. The steps taken to isolate, filter, and visualize data provided further insights regarding to the application and importance of time series methods, specifically in extracting and analyzing large real-life data with various levels of noise. As a final note, this project has successfully examined the different applications of time series analysis while estimating the orbital periods of selected exoplanets. The project demonstrates importance in adapting analysis techniques combined with physical knowledge, when managing ranges of different dataset, in order to extract reliable astrophysical insights.

6 References/Sources

^[1]W.Δ (2017, April 12). Exoplanet hunting in Deep Space. Kaggle.

<https://www.kaggle.com/datasets/keplersmachines/kepler-labelled-time-series-data>

^[2]Senad96 (2023, April 19). Senad96/exoplanet-detection: Find out if there are new planets using Deep Learning. GitHub.

<https://github.com/senad96/Exoplanet-Detection>

7 Appendix

7.1 Exo 4 & 5

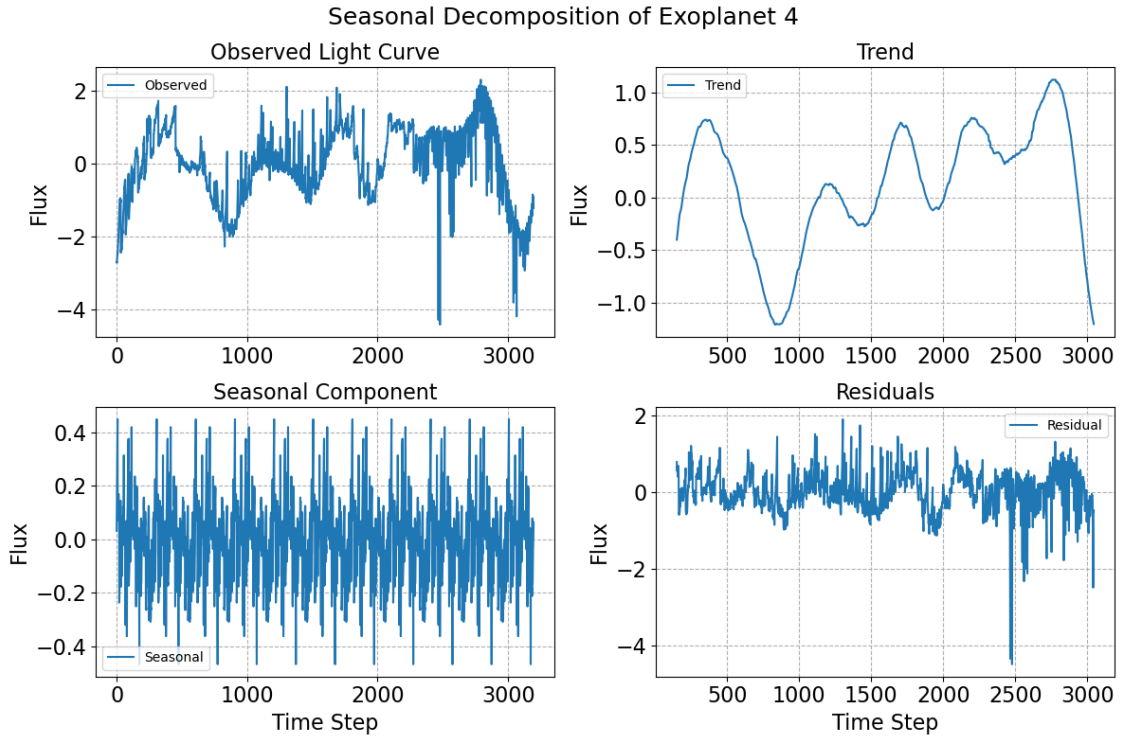


Figure 7: The seasonal decomposition plots of Exoplanet 4, with the observed light curve(Upper-Left), extracted trend(Upper-right), the seasonal component(lower-left), and the left residuals(Lower-right), all plotted with flux against time steps.

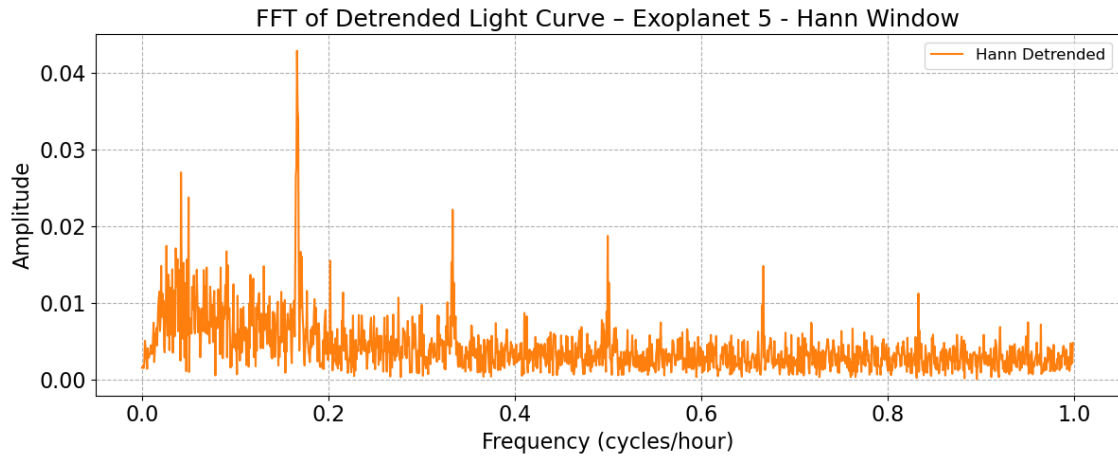


Figure 8: The Hann window filtered light curve in frequency domain of exoplanet 5, plotted with amplitude against frequency in cycles/hour, examined in supporting the concluding analysis of exoplanet 5.

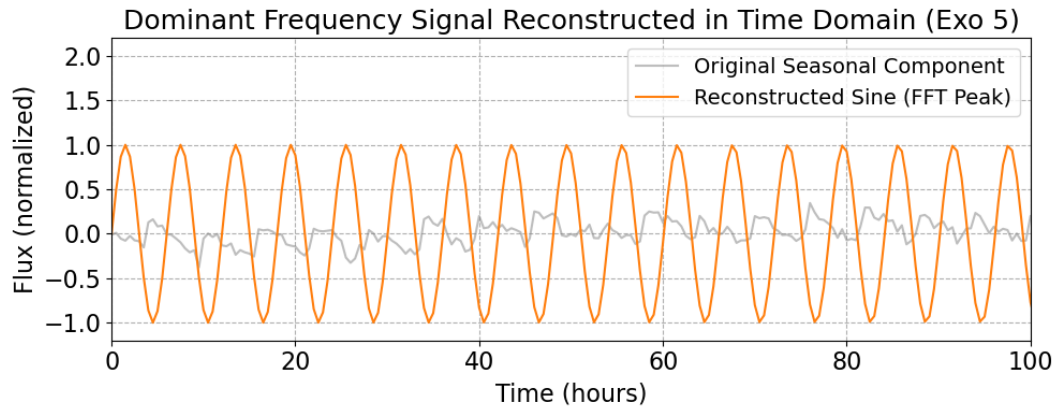


Figure 9: The Time-domain signal of exoplanet 5, reconstructed from the dominant frequency in frequency domain, plotted with normalized flux against time in hours, further supporting our period estimation of exoplanet 5.

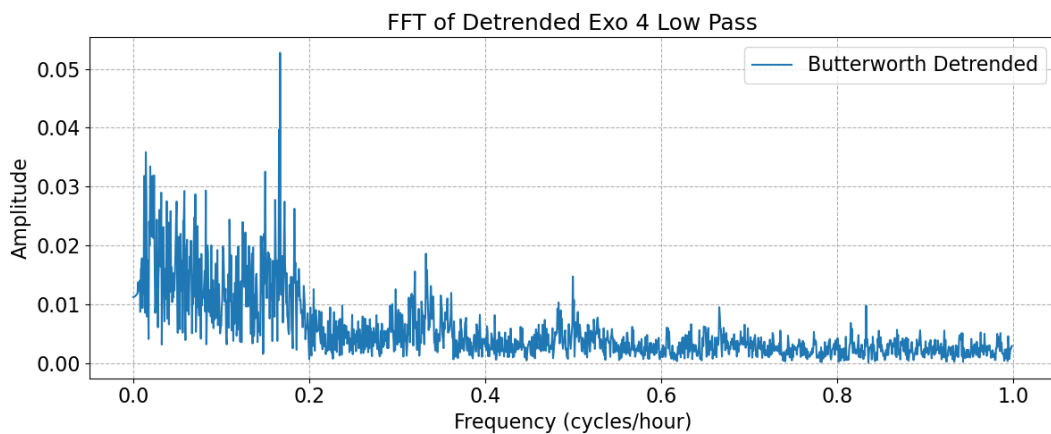


Figure 10: The graph of exoplanet 4' light curve in frequency domain after applying a low pass filter (Butterworth Detrended), plotted as amplitude against Frequency in cycles/hour, with some visibility in the dominant frequencies.

7.2 Exo 6

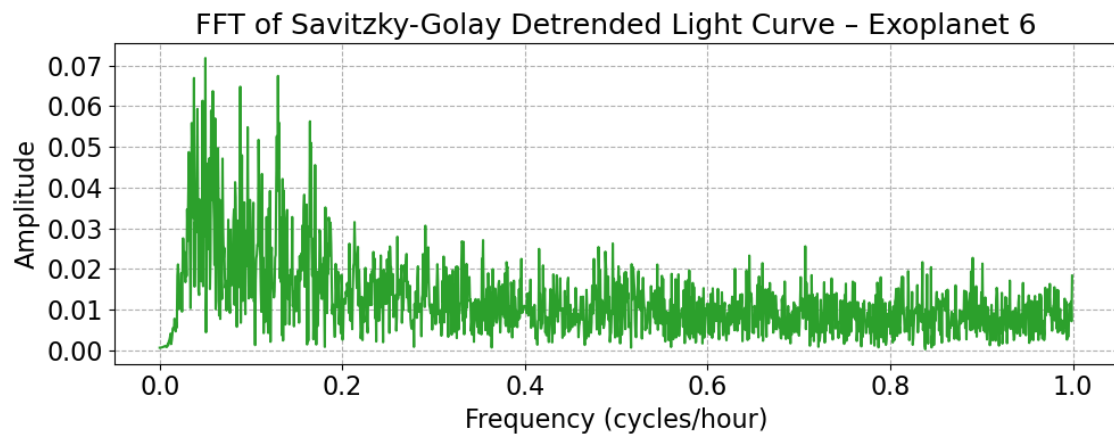


Figure 11: The plot of exoplanet 6's light curve in frequency domain after applying a Savitzky-Golay filter, with amplitude plotted against frequency in cycles/hour, displaying no clear peaks of analysis.

7.3 Exo 7

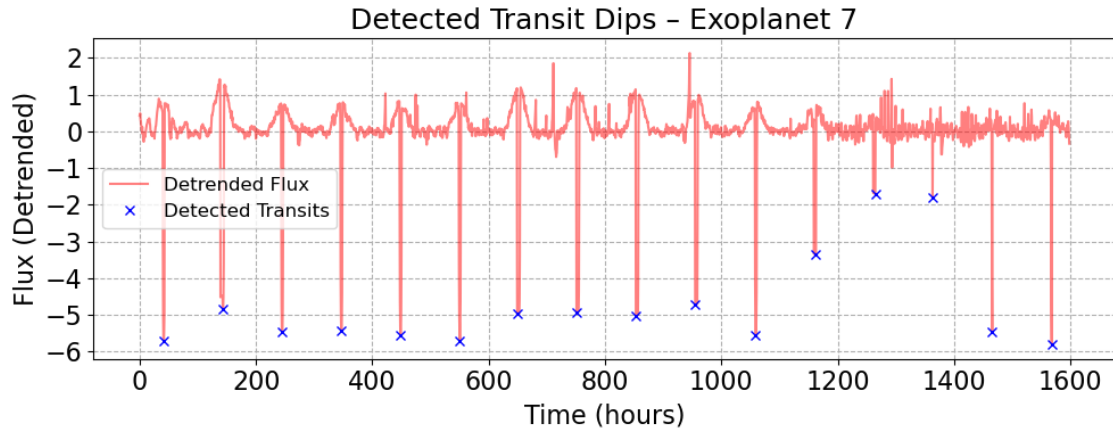


Figure 12: The visualization of peak detection method used in analyzing the correct period of exoplanet 7, with the detrended flux plotted against time in hours. The detected transits(assumed) are labeled in blue, also suggesting a minor anomaly around $t=1300$ hours.

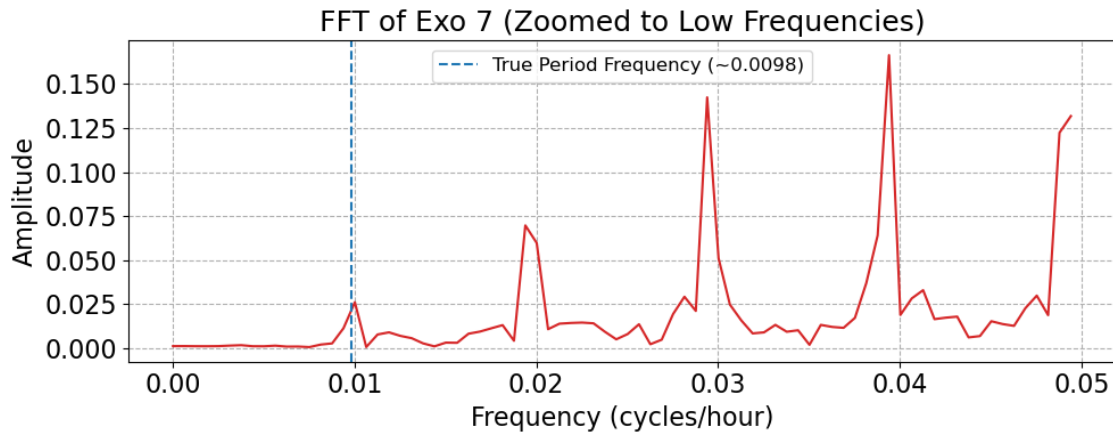


Figure 13: The frequency domain data of exoplanet 7 when zoomed in to low frequencies in search of the true period, with amplitude plotted against frequency in cycles/hour.

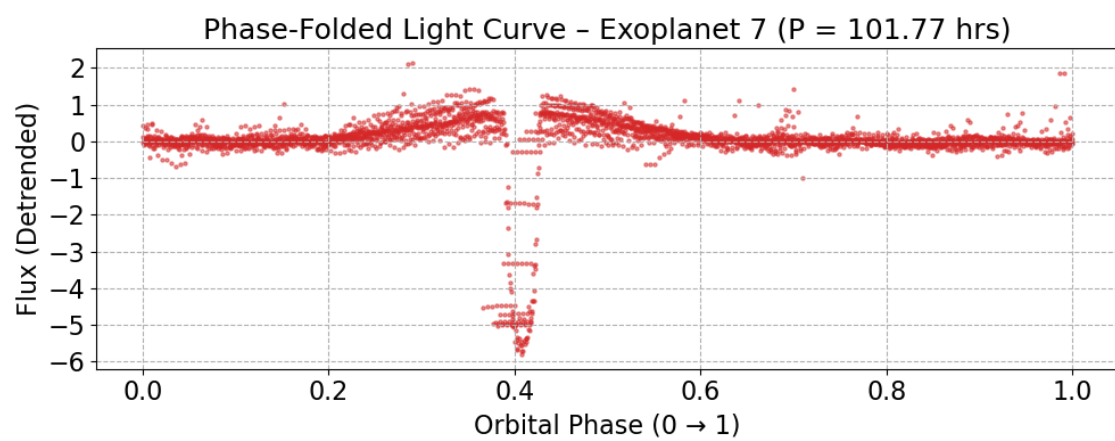


Figure 14: The plot of exoplanet's light curves being fold in the time domain, used for examining the reliability of the assumed results.